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MAM1019H

Hand-in 3

Date: August 15, 2022

Answers

1. (a) True
(b) False
(c) False
(d) True
2. This statement is false. We will prove that $\mathbb{R}$ and $\mathcal{P}(\mathbb{R})$ are both uncountable, but have different cardinalities. We know that there is no bijection $\mathbb{N} \rightarrow \mathbb{R}$, so $\mathbb{R}$ is uncountable. Note that $|A| < |\mathcal{P}(A)|$ for every set A , so $|\mathbb{R}| < |\mathcal{P}(\mathbb{R})|$ meaning $\mathcal{P}(\mathbb{R})$ is also uncountable and $|\mathbb{R}| \neq |\mathcal{P}(\mathbb{R})|$, thus not all uncountable sets have the same cardinality
3. (a) Since if $|A| = |B|$ for some sets A, B , then $|\mathcal{P}(A)| = |\mathcal{P}(B)|$, it is sufficient to prove that $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$, so then $|\mathcal{P}(\mathbb{N})| = |\mathcal{P}(\mathbb{N} \times \mathbb{N})|$.
Consider the table:

	1	2	3	4	...
1	(1,1)	(1,2)	(1,3)	(1,4)	...
2	(2,1)	(2,2)	(2,3)	(2,4)	...
3	(3,1)	(3,2)	(3,3)	(3,4)	...
4	(4,1)	(4,2)	(4,3)	(4,4)	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Now we can list the elements in the depicted way:

$(1,1), (1,2), (2,1), (3,1), (2,2), (1,3), (1,4), (2,3), (3,2), (4,1) \dots$

This list covers all of $\mathbb{N} \times \mathbb{N}$ with no repeated elements and so $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$ and thus this statement is true

- (b) First note that $|\mathcal{P}(\mathbb{N})| = |\mathbb{R}| = |(0,1)|$, so now what is left to prove is that $|(0,1) \times (0,1)| = |(0,1)|$. First note that any $x \in (0,1)$ has the form $x = 0.x_1x_2x_3x_4\dots$ where each x_i is the i th decimal place of x and $x_i \in \{0, 1, 2, \dots, 9\}$. Now define $f : (0,1) \times (0,1) \rightarrow (0,1)$ as $f(a,b) = 0.a_1b_1a_2b_2a_3b_3\dots$. We prove injectivity, so suppose $(a,b) \neq (a',b'), \forall (a,b), (a',b') \in (0,1) \times (0,1)$. Thus $a \neq a'$ or $b \neq b'$. Now, without loss of generality, assume that $a \neq a'$. This means $\exists n \in \mathbb{N}$ such that $a_n \neq a'_n$ (with a_n the n th decimal place of a). Notice a_n is the $(2n-1)$ th decimal place of $f(a,b)$ and a'_n is the $(2n-1)$ th decimal place of $f(a',b')$ and these differ, so $f(a,b) \neq f(a',b')$ and thus f is injective. Now consider the map $g : (0,1) \rightarrow (0,1) \times (0,1)$ defined as $g(x) = (x,x)$ for any

$x \in (0, 1)$. This is well defined because $x \in (0, 1) \Rightarrow (x, x) \in (0, 1) \times (0, 1)$. Also note that $g(x) = g(x') \Rightarrow (x, x) = (x', x') \Rightarrow x = x'$ for any $x, x' \in (0, 1)$. Thus g is injective. Now since we have an injection $(0, 1) \times (0, 1) \rightarrow (0, 1)$ and an injection $(0, 1) \rightarrow (0, 1) \times (0, 1)$, by CSB, we have that $|(0, 1)| = |(0, 1) \times (0, 1)|$ finally showing that $|\mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N})| = |(0, 1)|$